



CS-467- Software Project Management

Lecture No. 4

Software Scope Management

List of Topic to be Discuss in this lecture

- What is Project Scope Management?
- The Triple Constraints of Project Management
- The 6 project constraints
- Product and Project Scope
- Collect Requirements Process
- Define Scope Process
- Create Work Break Down Process
- Control Scope Process
- Validate Scope Process

Knowledge Area -Scope Management

Knowledge Area	Process				
	Initiating	Planning	Executing	Monitoring & Control	Closing
Scope Management		Collect Requirements Define Scope Create WBS		Control Scope Validate Scope	

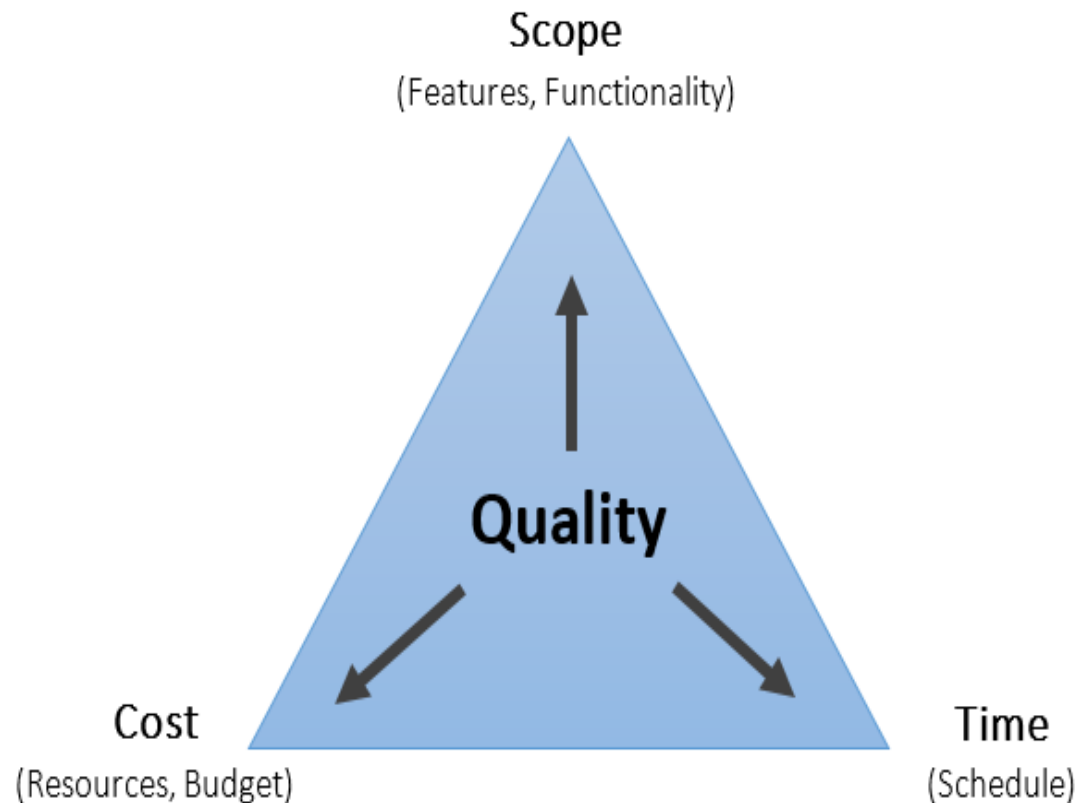
What is Project Scope Management?

Project scope management is a process that helps in determining and documenting the list of all the project goals, tasks, deliverables, deadlines, and budgets as a part of the planning process. Project objectives are listed below.

- **Meet user requirements:** Develop the project according to the user requirements after understanding them.
- **Meet schedule deadlines:** Complete the project milestones as described in the project plan on time in order to complete the project according to the schedule.
- **Be within budget:** Manage the overall project cost so that the project is within the allocated budget.
- **Produce quality deliverables:** Ensure that quality is considered for accuracy and overall performance of the project.

The Triple Constraints of Project Management

The triple constraints of project management—also known as the project management triangle or the iron triangle—are scope, cost, and time. You'll need to balance these three elements in every project, and doing so can be challenging because they all affect one another.



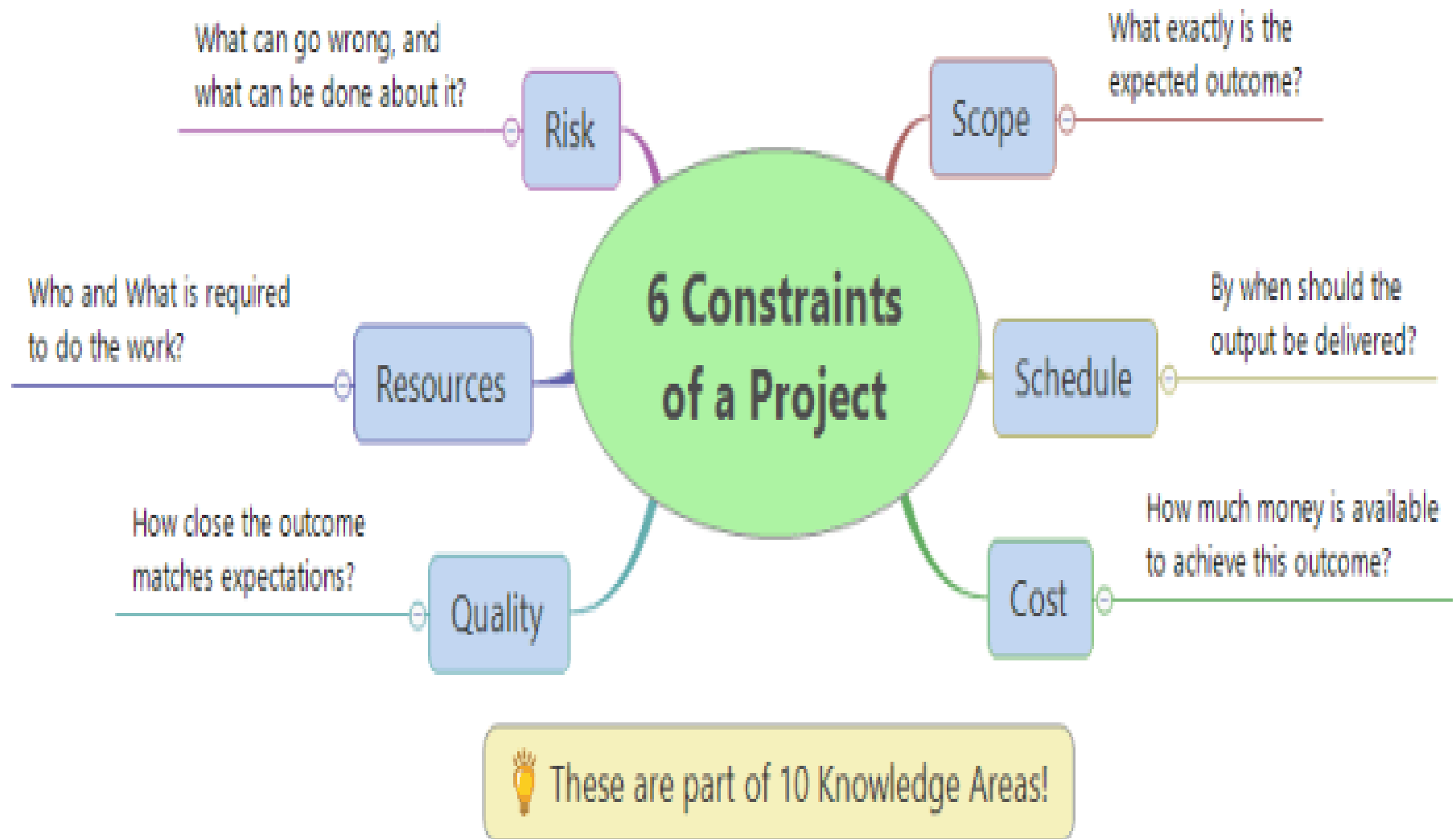
The Triple Constraints of Project Management

Time constraint: The time constraint refers to the project's schedule for completion, including the deadlines for each phase of the project, as well as the date for rollout of the final deliverable.

Scope constraint: The scope of a project defines its specific goals, deliverables, features, and functions, in addition to the tasks required to complete the project.

Cost constraint: The cost of the project, often dubbed the project's budget, comprises all of the financial resources needed to complete the project on time, in its predetermined scope. Keep in mind that cost does not just mean money for materials — it encompasses costs for labor, vendors, quality control, and other factors, as well.

The 6 project constraints



Some examples of how constraints are related

- ❑ If necessary resources are not available, time to deliver will increase. This may also increase project cost, because alternate resources if available, may be more expensive than planned.
- ❑ If QA team finds that quality of deliverable is going bad, more resources may be required. This increases the cost (additional resources) and effort to fix the faulty deliverable. This will also increase the time to deliver.
- ❑ If scope creep happens on the project, it will result in increased time, cost, resources and potentially reduced quality. And thus increased risk on delivery.

Product and Project Scope

Product Scope

Product scope identifies the characteristics and functions of a product or service. These characteristics include physical features such as size and materials, as well as functional specifications. Functional considerations include what the product is designed to do and its purpose or end-use. For example, if the product is a bridge, the product scope might be its length, width, and load strength.

Project Scope

Project scope is the work that must be done in order to deliver a product according to the product's scope (required functions and features). For example, a new telephone system would generally include four subsidiary components—hardware, software, training, and implementation.

- Completion of the project scope is measured against the project plan, but completion of the product scope is measured against the product requirements.
- Both types of scope management must be well integrated to ensure that the work of the project will result in delivery of the specified product.

1- Collect Requirements

- Collect requirements in the process in which requirements are collected from stakeholders to meet project objectives.
- These requirements will form the basis for the project scope and the work to be done on the project to make it successful.
- You need to know as many requirements as possible on the project to ensure project success, to avoid negative risk, and to take advantage of positive risk.

Tools and Techniques for Collecting Requirement

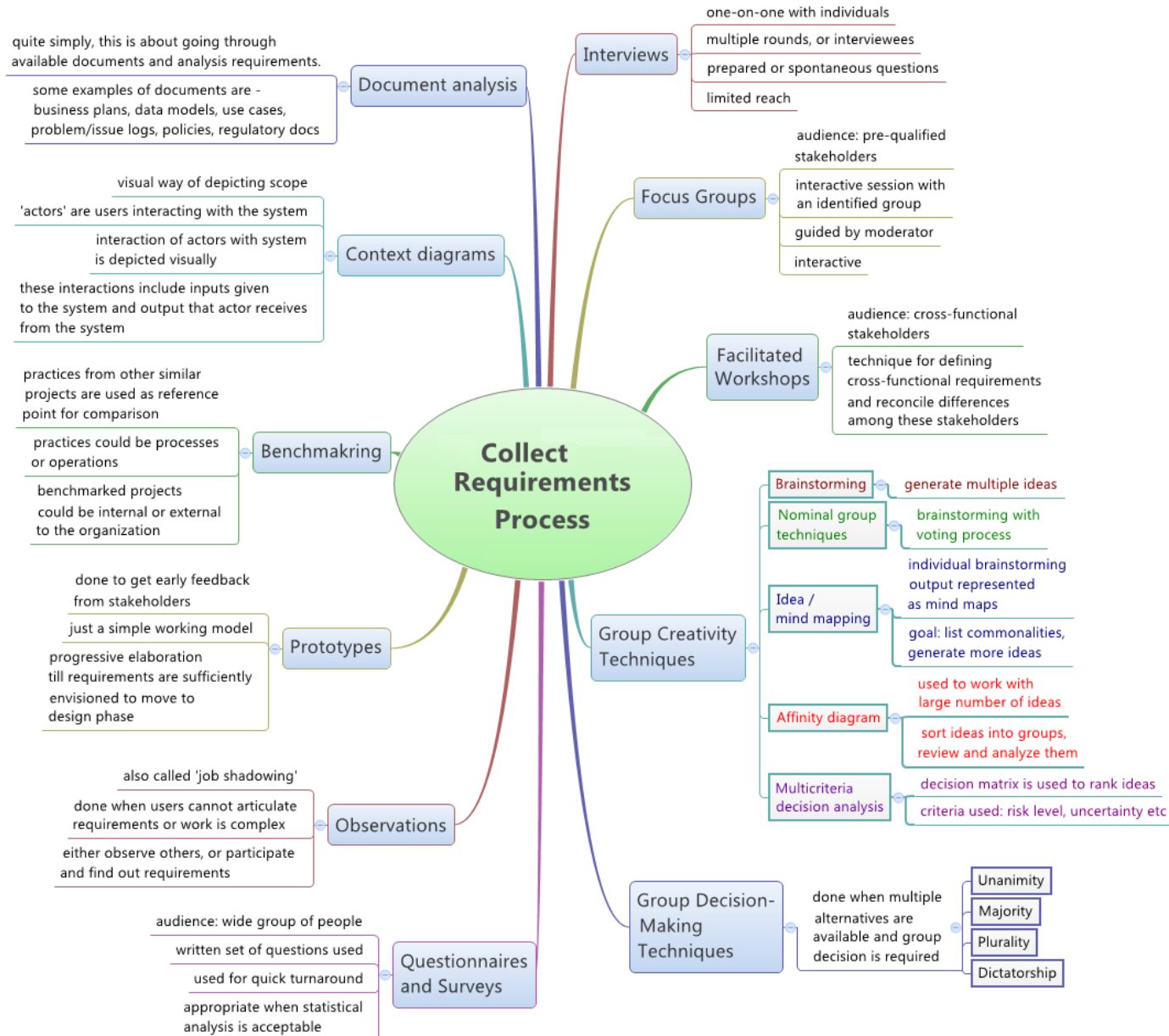
Interviews/Expert Interviews—Interviews are an approach for getting information from stakeholders by talking to them directly. The project manager interviews stakeholders to elaborate and explain their requirements on the project. Interviews may take place face to face, in groups, email, video chat etc.

Focus Groups—Focus groups include Getting expert opinion and requirements from the **same segment** of stakeholders and subject matter experts on an aspect of the project. For instance, talking to the Marketing Department to share their ideas on what the product design should look like. Focus group conversations are usually managed by a moderator.

Historical Records—Historical Records provide information on similar projects from the past which might have had similar requirements including product requirements, process requirements, quality requirements etc.

Facilitated Workshops—Stakeholders from **different segments** and viewpoints are brought together to define requirements.

Tools and Techniques for Collecting Requirement



Group Decisions Making Techniques

A big project usually has a lot of stakeholders, and that means a lot of opinions. You'll need to find a way of making decisions when those opinions conflict with each other. There are four major decision-making techniques you can choose from. These are referred to as group decision-making techniques on the test.

- **Unanimity** means everyone agrees on the decision.
- **Majority** means that more than half the people in the group agree on the decision.
- **Plurality** means that the idea that gets the most votes wins.
- **Dictatorship** is when one person makes the decision for the whole group.

Functional and Nonfunctional Requirements

The requirements document needs to list all of the functional and nonfunctional requirements of your product.

Functional requirements are most of the kinds of things that you think of right away: new features, bug fixes, and new or different behavior.

Nonfunctional requirements are sometimes called quality attributes because they're things that you expect from your deliverables, but aren't specific features.

Some examples of nonfunctional requirements are performance, reliability, error handling, and ease of use.

Requirements Document

The outputs of the Collect Requirements process are the **requirements document** and a **requirements traceability matrix**, which allows you to follow the requirements from the document through implementation and verification

REQUIREMENTS DOCUMENT

1. Introduction
2. Organization Impact
3. Functional requirements

ID:	FR-001
Description:	

4. Nonfunctional requirements

ID:	NFR-001
Description:	

Requirements Traceability Matrix

REQUIREMENTS TRACEABILITY MATRIX

<u>Req ID</u>	Priority	Test Case ID	Status
FR-001	High	TC-2	Approved

2- Define Scope Process

Define Scope is the process in which a **detailed description** of the project and product is developed. It includes important information including what is **included** in the project and what is **not included** in the project.

After Determining **requirements** and **Defining scope** on the project, the Project Manager can then determine the **budget** and **schedule**.

- **Scope** is **balanced** against the **schedule**, **budget**, and other **constraints**. The Project Manager needs to **adjust** the project to meet the **schedule** and **budget**.
- The objective is to develop a **realistic budget** and **schedule** that can **achieve the scope** without **conflicts** or **problems** .It is also important to resolve problems **before** actual **work begins on the project**.
- An **unrealistic budget** and **schedule** are a **project manager's fault**.
- **Output of define scope process** is **project scope statement**.

Project Scope Statement

PROJECT SCOPE STATEMENT

Product Scope Description: The product must contain 34 levels and 4 playable characters, and must be created for both Mac and PC platforms.

Project Exclusions: This project does not include a companion website. That will need to be done by another project team.

Project Deliverables: The deliverables for this project are:
Game, Test plan, Source code, Schedule, Design documents, Test reports, Defect reports, Change requests, Contract, Budget, Project Management plan

Project Acceptance Criteria: The product must not have an adverse impact on existing systems. All defects found must be judged of low enough priority and severity to be acceptable to all stakeholders.

Project Constraints: Artwork from the previous games cannot be used.

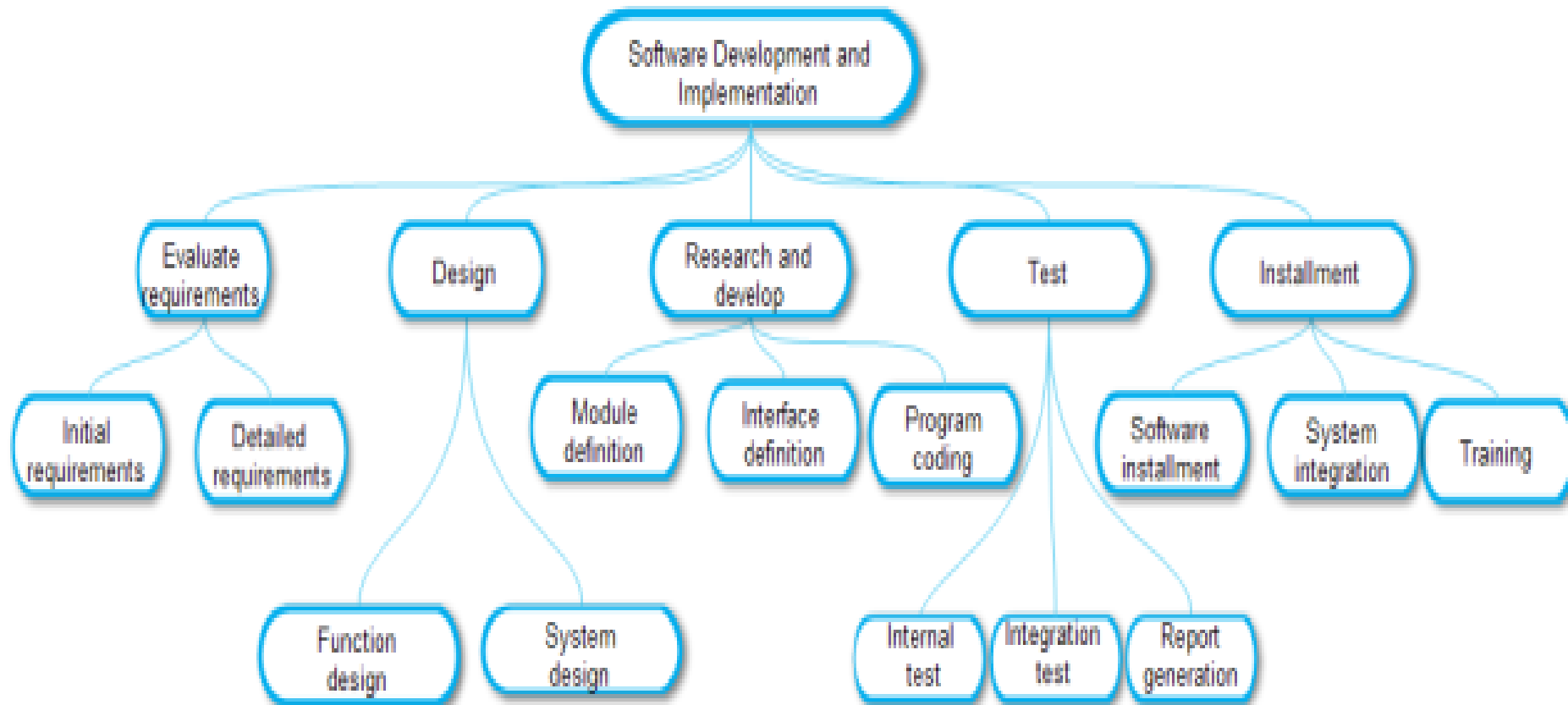
Project Assumptions: The developers will not be asked to work on any other projects

Create Work Break Down Process

- The **Create WBS** process is the most important process in the Scope Management knowledge area because it's where you actually figure out all the work you're going to do. It's where you create the **work breakdown structure** (or WBS), which is the main Scope Management output is shown as below Figure. **Collect Requirements and Define Scope become inputs to the Create WBS process.**



Create Work Break Down Process



4- Control Scope Process

The Anatomy of a change Control Scope Process

1. A change is needed.
2. Create a change request.
3. Get the change approved
4. Do variance analysis.
5. Re plans the work.
6. Create a new baseline.

Creating a Change Control System

You must create an effective change control system because changes are inevitable and even at times necessary.

The purpose of the change control system is to identify, monitor, and learn from the changes occurring in your project. Your change control system may consist of tools, such as a database or spreadsheet to record your proposed changes, and you may also have a ***change control board*** (CCB) to review and approve changes in your firm. Project team members would document their proposed changes and present them to the CCB for evaluation and approval.

If you do have a CCB, then all project changes would need to go before the CCB. The CCB might have just two members or it could be a large committee. The number of members varies from firm to firm. If you have too many members on your CCB, though, making decisions may take longer.

Change Request Form

Change Request Form

1. Submitter – General information

CR#: 001	Description of Request:
Requestor:	Reason for Change:
Date submitted:	
Date required:	Business Justification:
Priority:	
Expected Results:	

2. Change Review Board - Decision

Decision	Approved	Rejected	More info
Decision Date:	Signature:		

5- Validate Scope Process

- Validate Scope refers to **frequent meetings** with the **customer / sponsor** to **gain formal acceptance of deliverables**.
- The validate scope process occurs during monitoring and controlling process group.
- The Validate scope process is done **after receiving the deliverables** from the control quality process. During the control quality process, the deliverables are **checked and verified** from your end. After that, the deliverables are put in front of the **client** during validate scope process to gain **formal acceptance**.
- The customer will either accept the deliverables or request a change to be made. The important and major **outputs of Validate scope process** is **accepted deliverables**.

Validate Scope Form

VALIDATE SCOPE FORM

Deliverable Name	<Description of the deliverable to be accepted. These should be from the Scope Statement.>
Acceptance Criteria	<The criteria against which the deliverable will be judged>
Verification Method	<How will acceptance be verified by the project team?>
Validation Method	<How will acceptance be validated by the client (i.e. sponsor, customer, user acceptance group?)>
Client Name	<The name of the person responsible on the client's end for validating and accepting the deliverable>
Client Signature	
Signature Date	YYYY-MM-DD